

ITMO-444

Tool Set Up

Jay Han

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Lab1

Screenshot of your IAM dashboard showing “MFA on root user: Enabled” (blur/hide any sensitive info)

IAM Dashboard [Info](#)

Security recommendations 0

Root user has MFA
Having multi-factor authentication (MFA) for the root user improves security for this account.

Root user has no active access keys
Using access keys attached to an IAM user instead of the root user improves security.

IAM resources

Resources in this AWS Account

User groups	Users	Roles	Policies	Identity providers
0	1	3	0	0

What's new

Updates for features in IAM

- [AWS IAM now provides higher maximum quotas for roles, role trust policies, instance profiles, managed policies, and identity providers.](#) *3 months ago*
- [IAM Roles Anywhere now enforces VPC endpoint policies for the CreateSession API.](#) *3 months ago*
- [IAM Roles Anywhere now supports post-quantum digital certificates.](#) *5 months ago*
- [AWS simplifies IAM role creation and setup in service workflows.](#) *5 months ago*

more

[View all](#)

Screenshot of your IAM Users list showing your lab user

IAM users (1) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Q Search

< 1 > ⚙

☐

User name

▲

▼

Path

▼

Group

▼

Last activity

▼

MFA

▼

Password age

▼

Console last sign-in

▼

Access key ID

▼

Ac

☐

[Jay_Han](#)

/

0

✔

32 minutes ago

Virtual...

✔

25 minutes

✔

32 minutes ago

Active - AKIA4AN7ED...

✔

Screenshot showing MFA enabled on the IAM user

IAM Dashboard [Info](#)

Security recommendations 0

✔

Root user has MFA

Having multi-factor authentication (MFA) for the root user improves security for this account.

✔

You have MFA

Having multi-factor authentication (MFA) for the IAM user improves security for this account.

✔

Your user, Jay_Han, does not have any active access keys that have been unused for more than a year.

Deactivating or deleting unused access keys improves security.

IAM resources

Resources in this AWS Account

User groups

0

Users

1

Roles

3

Policies

0

Identity providers

0

Confirmation (text only — do not include actual key values) that you successfully generated an access key, e.g., AKIA****...**** (last 4 chars: XXXX)”

Access keys (1) [Create access key](#)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

AKIA

Description

itmo-444-lab

Last used

None

Last used region

N/A

Status

✔ Active

Created

23 minutes ago

Last used service

N/A

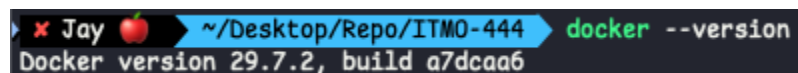
Actions

One paragraph reflecting on why root user MFA and least-privilege IAM users matter for cloud security.

Both the root user MFA as well as least-privilege IAM users provide layered security solutions for cloud environments by ensuring the separation of complete account control from actual operations. Given that the root user has god-level access, allowing the deletion of the whole cloud infrastructure or abusing the billing of accounts, protecting it with MFA guarantees that the compromise of only a single credential will not lead to complete account takeover. Furthermore, performing regular tasks with the help of IAM users with minimum rights reduces the area of damage in case of a security incident. In case a developer key or IAM password is exposed, the scope of attack will be restricted only to the services engaged, and there will be no possibilities for lateral movement.

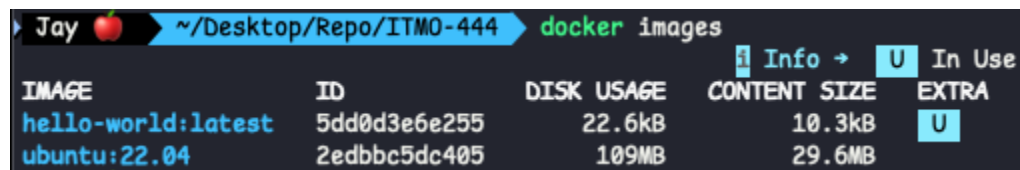
Lab2

Screenshot of docker --version output



```
➤ x Jay 🍏 ~/Desktop/Repo/ITMO-444 ➤ docker --version
Docker version 29.7.2, build a7dcaa6
```

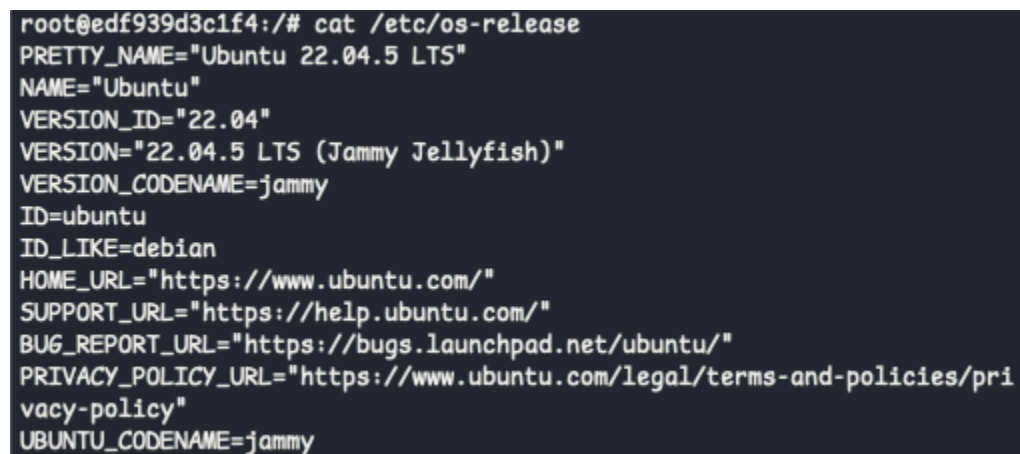
Screenshot of the hello-world container output



```
➤ Jay 🍏 ~/Desktop/Repo/ITMO-444 ➤ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5dd0d3e6e255	22.6kB	10.3kB	U
ubuntu:22.04	2edbbc5dc405	109MB	29.6MB	U

Screenshot of cat /etc/os-release run inside the Ubuntu container



```
root@edf939d3c1f4:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```

Screenshot of docker ps -a before and after removing my-ubuntu

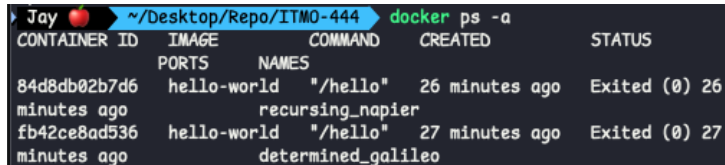
- Before removing my-ubuntu



A terminal window showing the command 'docker ps -a' and its output. The output lists four containers: 'edf939d3c1f4' (ubuntu:22.04, exited 127), '84d8db02b7d6' (hello-world, exited 0), 'fb42ce8ad536' (hello-world, exited 0), and 'determined_galileo' (determined_galileo, exited 0). The 'my-ubuntu' container is not present in the list.

CONTAINER ID	IMAGE	PORTS	NAMES	COMMAND	CREATED	STATUS
edf939d3c1f4	ubuntu:22.04			"/bin/bash"	2 minutes ago	Exited (127) 9 seconds ago
84d8db02b7d6	hello-world			"/hello"	25 minutes ago	Exited (0) 25 minutes ago
fb42ce8ad536	hello-world			"/hello"	26 minutes ago	Exited (0) 26 minutes ago
determined_galileo	determined_galileo					

- After removing my-ubuntu



A terminal window showing the command 'docker ps -a' and its output after removing the 'my-ubuntu' container. The output now lists only three containers: '84d8db02b7d6' (hello-world, exited 0), 'fb42ce8ad536' (hello-world, exited 0), and 'determined_galileo' (determined_galileo, exited 0). The 'my-ubuntu' container is no longer present.

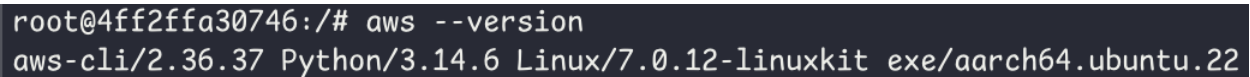
CONTAINER ID	IMAGE	PORTS	NAMES	COMMAND	CREATED	STATUS
84d8db02b7d6	hello-world			"/hello"	26 minutes ago	Exited (0) 26 minutes ago
fb42ce8ad536	hello-world			"/hello"	27 minutes ago	Exited (0) 27 minutes ago
determined_galileo	determined_galileo					

One paragraph explaining, in your own words, the difference between a Docker image and a Docker container

A Docker image is a type of image that is both static and immutable, containing an application with its dependencies. While a Docker container is the running form of this image and application, it is independent from any other processes running on the computer. It's important to realize that the image acts as the blueprint to share, whereas the container, being the running application, represents a new instance of it, with additional information about its execution.

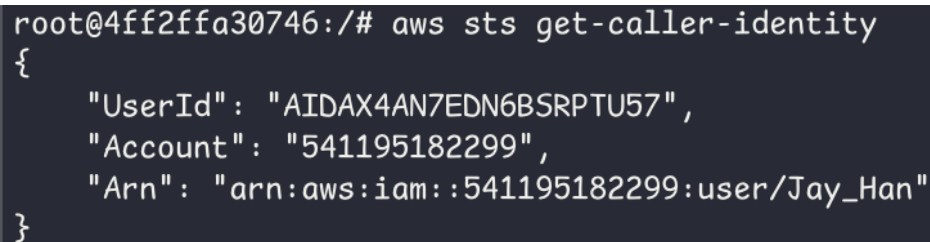
Lab3

Screenshot of aws --version output.



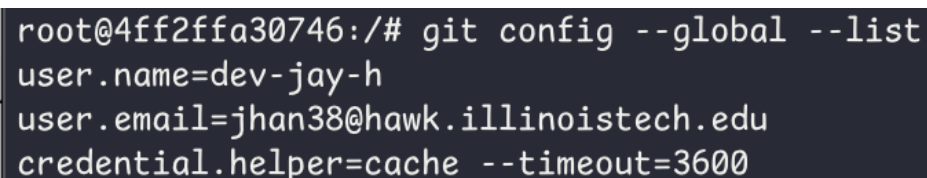
```
root@4ff2ffa30746:/# aws --version
aws-cli/2.36.37 Python/3.14.6 Linux/7.0.12-linuxkit exe/aarch64.ubuntu.22
```

Screenshot of aws sts get-caller-identity showing your IAM username in the ARN (not root).



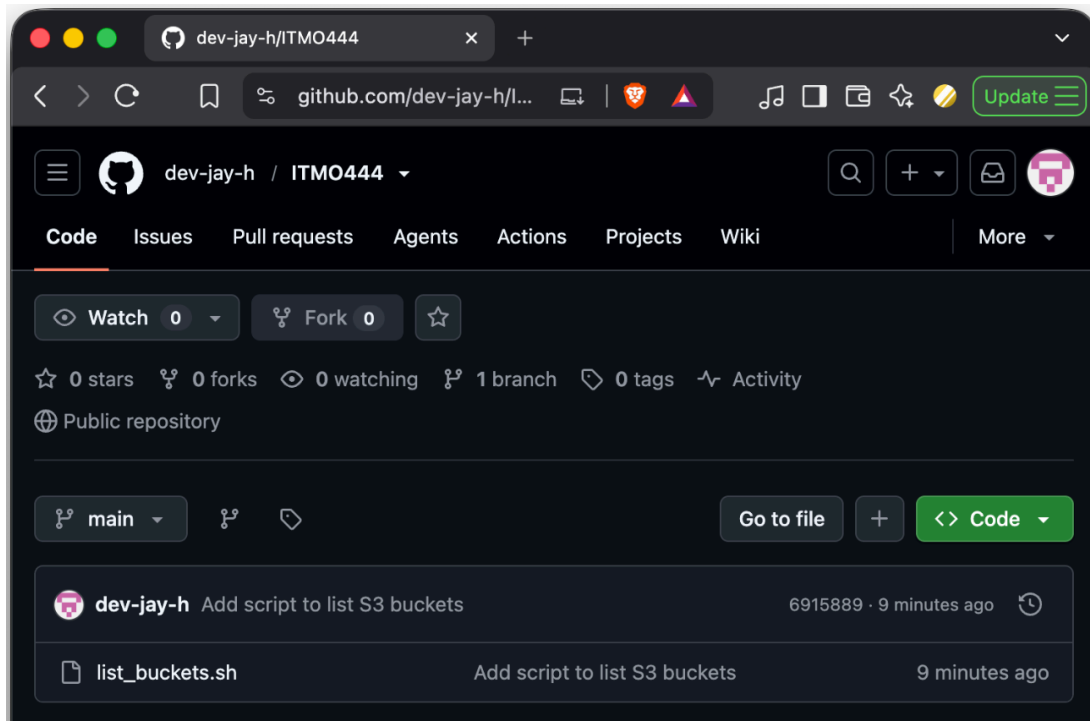
```
root@4ff2ffa30746:/# aws sts get-caller-identity
{
  "UserId": "AIDAX4AN7EDN6BSRPTU57",
  "Account": "541195182299",
  "Arn": "arn:aws:iam::541195182299:user/Jay_Han"
}
```

Screenshot of git config --global --list showing your name and college email



```
root@4ff2ffa30746:/# git config --global --list
user.name=dev-jay-h
user.email=jhan38@hawk.illinoistech.edu
credential.helper=cache --timeout=3600
```

Screenshot of your GitHub repository showing list_buckets.sh successfully pushed.



One paragraph explaining why you should never commit AWS credentials or GitHub tokens to a repository, even a private one.

The storage of sensitive data is usually in Git repositories. Secret data such as AWS security credentials or personal tokens like GitHub passwords are recorded by Git in its history of events. Even though there are users who may think that storing those pieces of information is not dangerous in private repositories, this concept is not acceptable. The credentials can be stolen in situations like changes to collaborators' account credentials, publicizing a private repository by accident, allowing malicious third parties to access the project, etc. Once the credentials become known, a bot may be programmed to perform certain destructive actions, such as stealing AWS credentials in several minutes and causing serious harm to its owner.